Norris Phase Report 5.1

Phase 5 final: held-out evaluation consolidated, with exploratory robustness analyses

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Repository github.com/SHAZAN01/csudh-norris-ape-2026 (phase4-main, to be merged into main)

Phase 5 complete. The confirmatory results of Report 5 stand unchanged; ten exploratory analyses, run from committed evidence only, confirm them and sharpen two interpretations. One wording in Report 5 (RQ1) is corrected below, and three sentences of the manuscript were revised accordingly.

Executive summary

What this report adds. Report 5 delivered the frozen confirmatory analysis: two Holm families of 24 paired-bootstrap tests, harm bands, protect-pool over-editing, preservation screens and a 96-case source review. Report 5.1 asks how robust those conclusions are and where the effects come from. All analyses below are *exploratory*, declared after the confirmatory analysis, computed by a new script (*src/analysis/phase5_extended.py*) from *main_results.json* and the Phase 4 run summary, and verified by unit tests against SciPy and a published Newcombe example. No frozen file, threshold, family or research question was changed.

- **The confirmatory conclusions are robust.** No F1 decision changes under Bonferroni or Benjamini–Hochberg correction, or under Holm over both families jointly (48 tests). All 18 Holm-significant F1 effects have the same sign in COMET, BLEU and chrF++; across all 24 conditions the COMET deltas rank-correlate with chrF++ at $\rho = 0.97$ and with BLEU at $\rho = 0.94$.
- **Where the gains come from.** An exact decomposition of each mean ΔCOMET by baseline-quality pool reproduces all 24 F1 deltas (maximum error 2e-16 points). For GPT-4o Mini, weak drafts (S0 < 0.80, 6–18% of items) supply 41–77% of each gain; strong drafts move the mean by at most 0.044 points. For Llama, edits help weak drafts in 11 of 12 conditions, but losses on mid-quality and strong drafts outweigh them: even an oracle that left strong drafts untouched would keep Llama below S0 in 6 of 12 conditions.
- **RQ1 correction.** Reweighting each direction to a common quality mix narrows GPT-4o Mini's Tagalog–Spanish gap by only 16–23%; at least 73% of the remaining gap arises in mid-quality drafts. Report 5 §9 said the pair difference "tracks the share of weak drafts more than language"; that was too strong. Baseline quality explains part of the difference, not most of it. The manuscript now says so.
- **The over-editing gap is not a sampling artefact.** Wilson 95% intervals separate the pooled protect-pool rates (GPT-4o Mini at most 5.6%, Llama at least 9.5%), and the Llama − GPT-4o Mini difference excludes zero for every strategy at every sensitivity threshold. Per direction, the intervals separate out of English (En→Tl, En→Es) but overlap into English (Tl→En, Es→En).
- **Edit quality, not edit quantity.** Among edited drafts, every GPT-4o Mini condition harms at most 8.9%, below the 9.9% minimum for Llama. GPT-4o Mini makes 2.3–16.1 improvements per harm; Llama 0.31–1.47, below one in 9 of 12 conditions.
- **Strategy ranking and cost.** GPT-4o Mini ranks one-shot > zero-shot > reason-then-edit in all four directions (Kendall's W = 1.00); reason-then-edit is last for both editors everywhere. One-shot costs 31% more than zero-shot and yields a 36% larger mean gain; reason-then-edit costs 42% more and yields 19% less.

Check	Result
Inputs	main_results.json (evaluation aaff72ad3e0a432c…) and Phase 4 GPT run summary; no text, generations or GPU
Holm recomputation	F1 and F2 Holm p-values recomputed from raw p: maximum difference 0 and 0
Pool decomposition	Σ(pool share × pool mean) equals each of the 24 F1 deltas; maximum error 2.2e-16 points
Practical-band flag	Recomputed ±0.5 flags equal the frozen evaluator's flags for all 48 contrasts
Token accounting	Phase 5 token counts equal the Phase 4 run summary for all 12 GPT-4o Mini conditions
Statistics code	Wilson, Spearman, BH and Holm equal SciPy to 1e−12; Newcombe interval reproduces the published 0.0524–0.3339
Manuscript audit	77 held-out claims (66 + 11 new) verified against evidence; planted errors in new claims detected; 5 pages

1. Status of these analyses

The approved plan (17 October) allows analyses beyond the two confirmatory families if they are labelled exploratory and their multiplicity treatment is stated. Accordingly: (i) none of the analyses below adds a test to F1 or F2; (ii) where intervals are reported they are pointwise and descriptive, with no new significance claims; (iii) any quantity that uses the reference-based S0 COMET to decide which drafts to edit is labelled **oracle**, because a deployed system cannot see the reference; (iv) the analyses use condition-level aggregates committed in the repository. Item-level indicators remain private on Drive, so intervals for over-editing differences treat the two editors' samples as independent (Section 6).

The pool definitions are the frozen ones: *weak* (improve pool) S0 COMET < 0.80; *strong* (protect pool) S0 COMET ≥ 0.90; *middle* in between. Pool sizes per direction (weak / middle / strong): En→Tl 183 / 595 / 234; Tl→En 82 / 553 / 377; En→Es 104 / 633 / 275; Es→En 60 / 659 / 293.

2. Confirmatory results at a glance (unchanged from Report 5)

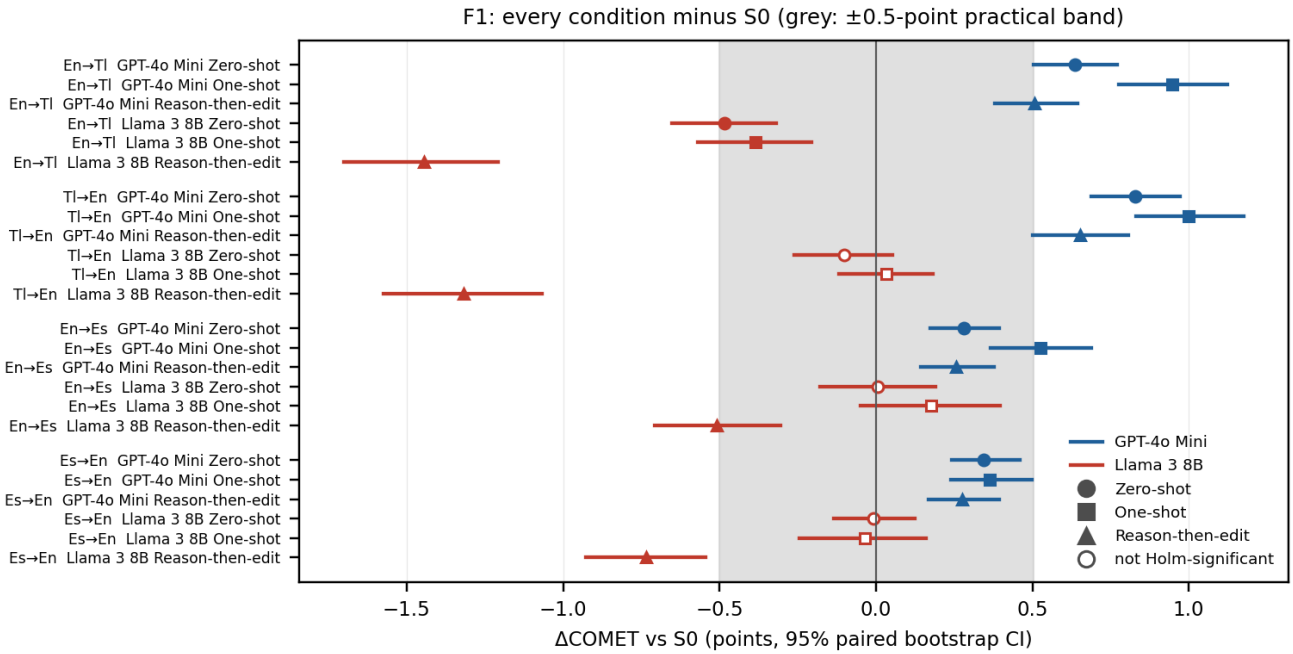


Figure 1. Primary family F1: each condition minus S0 with paired percentile 95% intervals (10,000 resamples, seed 42). Filled markers are significant after Holm correction over 24 tests; the grey band is the prespecified ±0.5-point practical band (a flag, not an equivalence test). Full F1/F2 tables with raw and Holm p-values are in Report 5, Sections 2–3.

GPT-4o Mini improves all twelve conditions (+0.26 to +1.00 points); Llama 3 8B improves none and is significantly worse in six. In the strategy family, 15 of 24 contrasts are significant: one-shot beats zero-shot for GPT-4o Mini in three directions, and reason-then-edit is never significantly better than either alternative.

3. Robustness of the inferential decisions

3.1 Alternative multiplicity corrections (X1)

Family (24 tests each)	Raw p < .05	Holm (frozen)	Bonferroni	Benjamini–Hochberg (FDR)	Holm over all 48
F1: condition − S0	18	18	18	18	18
F2: strategy contrasts	18	15	15	18	15

Number of significant contrasts at α = 0.05 under each correction. Holm within each family is the frozen procedure; the others are sensitivity checks only.

Every F1 decision is identical under all four procedures, and every F2 decision is identical under Bonferroni and under a single Holm family of 48. A less conservative FDR criterion would add three F2 contrasts, all small and none changing a conclusion: Llama 3 8B Tl→En one-shot - zero-shot +0.13 (raw p = 0.031); GPT-4o Mini En→Tl reason-then-edit - zero-shot −0.13 (raw p = 0.011); GPT-4o Mini Es→En reason-then-edit - zero-shot −0.07 (raw p = 0.034). The frozen Holm results are therefore, if anything, conservative. Resolution note: 17 of the 18 significant F1 tests and 14 of the 15 significant F2 tests sit at the bootstrap floor p = 1/10,001; with 24 tests the smallest attainable Holm value is 0.0024, comfortably below 0.05. The closest non-significant contrast is GPT-4o Mini En→Tl reason-then-edit − zero-shot (−0.13; raw p = 0.011, Holm p = 0.102).

3.2 Width of the practical band (X2)

Editor	±0.25 points: beyond / within / straddles	±0.5 points (frozen)	±1.0 point
GPT-4o Mini	7 / 0 / 5	5 / 5 / 2	0 / 10 / 2
Llama 3 8B	5 / 4 / 3	3 / 6 / 3	2 / 10 / 0

Counts of the 12 F1 intervals per editor that lie entirely beyond the band (|effect| larger than the band with 95% confidence), entirely within it, or straddle its edge.

Two readings are stable. First, no GPT-4o Mini interval lies beyond ±1 point: the gains are real but modest at every band width. Second, Llama’s reason-then-edit losses remain beyond the band in 2 directions even at ±1 point. At ±0.25 points, 7 of 12 GPT-4o Mini gains are clearly beyond the band, so “small” is not “negligible”.

4. Where gains and losses come from: exact pool decomposition (X3)

For each condition the mean ΔCOMET over 1,012 items equals Σ_{pool} (n_{pool} / 1,012) × mean $\Delta COMET_{pool}$. The three terms are the contributions of weak, middle and strong drafts. Because the identity is exact, it is also a cross-artifact check: the pool means from the evaluator's stratified summaries reproduce the F1 deltas from the bootstrap module to floating-point precision.

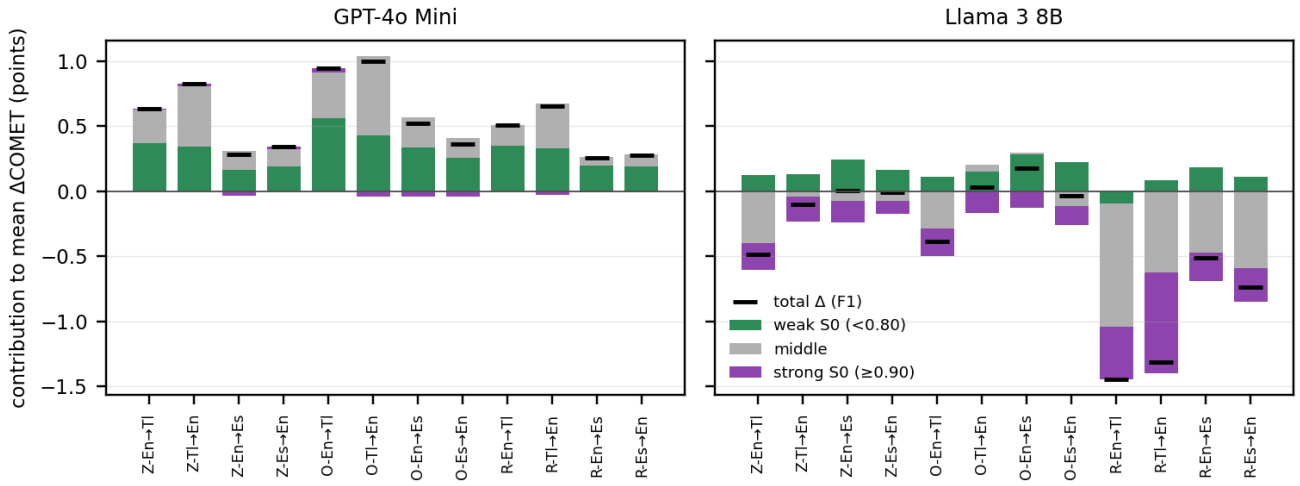


Figure 2. Contribution of each baseline-quality pool to the mean ΔCOMET (points). Bars stack positive and negative contributions separately; the black tick is the F1 total. Z/O/R = zero-shot / one-shot / reason-then-edit.

Editor	Strategy	Dir.	Weak	Middle	Strong	Total = F1	Oracle: weak only	Oracle: skip strong
GPT-4o Mini	Zero-shot	En→Tl	+0.370	+0.259	+0.007	+0.635	+0.370	+0.629
GPT-4o Mini	Zero-shot	Tl→En	+0.341	+0.466	+0.021	+0.828	+0.341	+0.807
GPT-4o Mini	Zero-shot	En→Es	+0.167	+0.146	−0.032	+0.282	+0.167	+0.314
GPT-4o Mini	Zero-shot	Es→En	+0.190	+0.137	+0.017	+0.344	+0.190	+0.327
GPT-4o Mini	One-shot	En→Tl	+0.563	+0.350	+0.034	+0.947	+0.563	+0.913
GPT-4o Mini	One-shot	Tl→En	+0.429	+0.609	−0.038	+1.000	+0.429	+1.038
GPT-4o Mini	One-shot	En→Es	+0.338	+0.231	−0.044	+0.525	+0.338	+0.569
GPT-4o Mini	One-shot	Es→En	+0.256	+0.151	−0.044	+0.364	+0.256	+0.407
GPT-4o Mini	Reason-then-edit	En→Tl	+0.349	+0.161	−0.002	+0.507	+0.349	+0.509
GPT-4o Mini	Reason-then-edit	Tl→En	+0.328	+0.349	−0.024	+0.653	+0.328	+0.677
GPT-4o Mini	Reason-then-edit	En→Es	+0.196	+0.067	−0.008	+0.255	+0.196	+0.263
GPT-4o Mini	Reason-then-edit	Es→En	+0.194	+0.088	−0.007	+0.275	+0.194	+0.282
Llama 3 8B	Zero-shot	En→Tl	+0.123	−0.397	−0.210	−0.484	+0.123	−0.274
Llama 3 8B	Zero-shot	Tl→En	+0.134	−0.040	−0.195	−0.100	+0.134	+0.094
Llama 3 8B	Zero-shot	En→Es	+0.247	−0.072	−0.171	+0.005	+0.247	+0.175
Llama 3 8B	Zero-shot	Es→En	+0.163	−0.074	−0.097	−0.007	+0.163	+0.089
Llama 3 8B	One-shot	En→Tl	+0.109	−0.287	−0.209	−0.386	+0.109	−0.177
Llama 3 8B	One-shot	Tl→En	+0.153	+0.049	−0.168	+0.034	+0.153	+0.202
Llama 3 8B	One-shot	En→Es	+0.282	+0.018	−0.125	+0.175	+0.282	+0.301
Llama 3 8B	One-shot	Es→En	+0.226	−0.111	−0.151	−0.035	+0.226	+0.116
Llama 3 8B	Reason-then-edit	En→Tl	−0.092	−0.950	−0.402	−1.444	−0.092	−1.042
Llama 3 8B	Reason-then-edit	Tl→En	+0.085	−0.625	−0.777	−1.318	+0.085	−0.540
Llama 3 8B	Reason-then-edit	En→Es	+0.183	−0.473	−0.218	−0.509	+0.183	−0.290
Llama 3 8B	Reason-then-edit	Es→En	+0.114	−0.595	−0.255	−0.736	+0.114	−0.481

Contributions in COMET points. Oracle columns: the mean change if edits were kept only for weak drafts, or for all but strong drafts, with S0 elsewhere. They use the reference-based S0 score and are not deployable; they show headroom, not an achievable system.

Reading. GPT-4o Mini's gains rest on weak and middle drafts, while strong drafts are essentially neutral (|contribution| ≤ 0.044). The weak pool is small (from 6% of items in Es→En to 18% in En→Tl) yet carries 41–77% of each GPT-4o Mini gain. Llama shows the opposite profile: its weak-pool contribution is positive in 11 of 12 conditions (all but reason-then-edit En→Tl), so the model can repair bad drafts, but its losses on mid-quality and strong drafts dominate. Leaving strong drafts untouched would still leave 6 Llama conditions below S0, all of them with negative middle-pool contributions. Protecting strong drafts is necessary but not sufficient: an open 8B editor would also need to leave adequate mid-quality drafts alone. This motivates quality-estimation gating as future work; it was not part of the frozen design and is not claimed.

5. RQ1 revisited: quality-standardized pair comparison (X4)

RQ1 asks how gains differ between the pairs *conditional on baseline quality*. Direct standardization answers this descriptively: each direction's pool means are reweighted to the pooled four-direction quality mix (weak 10.6%, middle 60.3%, strong 29.1%). If the Tagalog pair gained more only because it has more weak drafts, the standardized gap should vanish.

Editor	Strategy	En→Tl raw / std	Tl→En raw / std	En→Es raw / std	Es→En raw / std	Tl − Es gap raw	Gap std	Reduction
GPT-4o Mini	Zero-shot	+0.64 / +0.49	+0.83 / +0.98	+0.28 / +0.28	+0.34 / +0.48	+0.42	+0.35	16%
GPT-4o Mini	One-shot	+0.95 / +0.73	+1.00 / +1.20	+0.53 / +0.52	+0.36 / +0.55	+0.53	+0.43	19%
GPT-4o Mini	Reason-then-edit	+0.51 / +0.37	+0.65 / +0.79	+0.26 / +0.26	+0.27 / +0.42	+0.31	+0.24	23%
Llama 3 8B	Zero-shot	−0.48 / −0.60	−0.10 / −0.02	+0.00 / +0.00	−0.01 / +0.13	−0.29	−0.37	widens 29%
Llama 3 8B	One-shot	−0.39 / −0.49	+0.03 / +0.12	+0.18 / +0.17	−0.04 / +0.15	−0.25	−0.35	widens 41%
Llama 3 8B	Reason-then-edit	−1.44 / −1.53	−1.32 / −1.19	−0.51 / −0.50	−0.74 / −0.60	−0.76	−0.81	widens 6%

Raw = F1 delta; std = Σ pooled weight × direction pool mean. Gap = mean of the two Tagalog directions minus the mean of the two Spanish directions. For Llama the gap is negative (larger losses in the Tagalog pair) and standardization slightly widens it.

Correction to Report 5, Section 9 (RQ1). Report 5 stated that the pair difference "tracks the share of weak drafts more than language". Standardization narrows GPT-4o Mini's Tagalog–Spanish gap by only 16–23%, and 73–84% of the remaining gap comes from mid-quality drafts, where GPT-4o Mini gains more in the Tagalog pair. **Corrected reading:** baseline-quality composition explains part of the pair difference; the remainder may reflect finer quality differences inside the broad middle pool or genuine pair effects, which three coarse pools cannot separate, and resource level stays confounded with language, direction and domain. The manuscript sentence was revised to this reading (Section 10).

6. Uncertainty in over-editing (X5) and edit outcomes (X6)

6.1 Protect-pool over-editing with Wilson intervals

Editor	Strategy	Loss > 0.01	Loss > 0.02 (primary)	Loss > 0.05
GPT-4o Mini	Zero-shot	22/1179 = 1.9% [1.2, 2.8]	10/1179 = 0.8% [0.5, 1.6]	5/1179 = 0.4% [0.2, 1.0]
GPT-4o Mini	One-shot	140/1179 = 11.9% [10.2, 13.8]	51/1179 = 4.3% [3.3, 5.6]	8/1179 = 0.7% [0.3, 1.3]
GPT-4o Mini	Reason-then-edit	36/1179 = 3.1% [2.2, 4.2]	25/1179 = 2.1% [1.4, 3.1]	6/1179 = 0.5% [0.2, 1.1]
Llama 3 8B	Zero-shot	220/1179 = 18.7% [16.5, 21.0]	132/1179 = 11.2% [9.5, 13.1]	36/1179 = 3.1% [2.2, 4.2]
Llama 3 8B	One-shot	259/1179 = 22.0% [19.7, 24.4]	133/1179 = 11.3% [9.6, 13.2]	32/1179 = 2.7% [1.9, 3.8]
Llama 3 8B	Reason-then-edit	391/1179 = 33.2% [30.5, 35.9]	247/1179 = 20.9% [18.7, 23.4]	94/1179 = 8.0% [6.6, 9.7]
Llama − GPT	Same strategy	Z 16.8 [14.5, 19.2]; O 10.1 [7.1, 13.1]; R 30.1 [27.2, 33.0]	Z 10.3 [8.5, 12.3]; O 7.0 [4.8, 9.1]; R 18.8 [16.4, 21.3]	Z 2.6 [1.6, 3.8]; O 2.0 [1.0, 3.2]; R 7.5 [5.9, 9.2]

Pooled over the four directions (1,179 strong drafts). Brackets: Wilson 95% intervals; last row: Newcombe hybrid-score intervals for the difference in percentage points (Z/O/R), treating the editors as independent samples because item-level indicators are not in the committed evidence.

At every threshold and for every strategy the difference interval excludes zero (smallest lower bound 1.0 points). Per direction at the primary threshold, the highest GPT-4o Mini upper bound lies below the lowest Llama lower bound in En→Tl, En→Es, but the intervals overlap in Tl→En, Es→En, where the Llama zero-shot and one-shot rates come closest to GPT-4o Mini one-shot. The pooled ordering is secure; direction-level ordering into English should be stated

with that caveat.

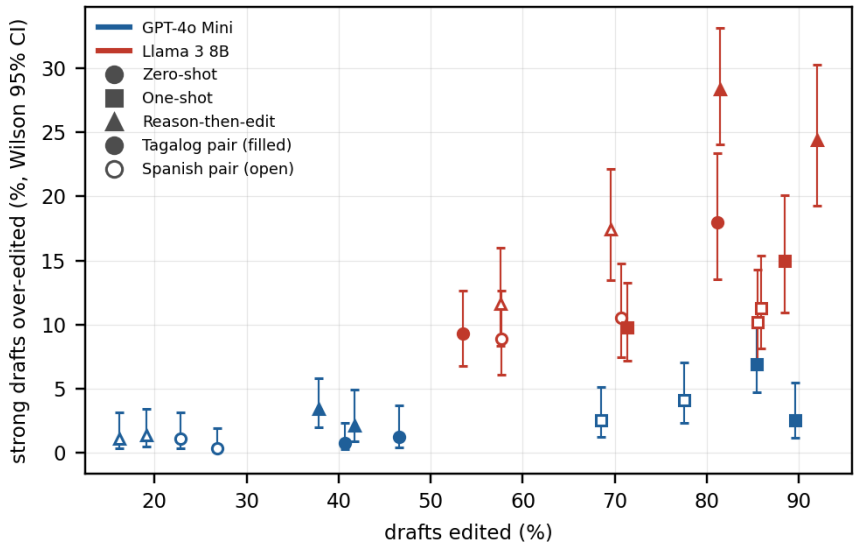


Figure 3. Share of drafts edited against the share of strong drafts over-edited (Wilson 95% intervals), one point per condition. Edit rates overlap between the editors (GPT-4o Mini one-shot edits 69–90%); over-editing rates do not.

6.2 What happens when an editor does change a draft

Editor	Strategy	Dir.	Edited	Improved / edited	Harmed / edited	Improved : harmed	Net per 1,000 items
GPT-4o Mini	Zero-shot	En→Tl	471	27.4%	3.8%	7.17	+110
GPT-4o Mini	Zero-shot	Tl→En	412	35.2%	2.2%	16.11	+134
GPT-4o Mini	Zero-shot	En→Es	231	22.9%	5.6%	4.08	+40
GPT-4o Mini	Zero-shot	Es→En	271	20.7%	3.3%	6.22	+46
GPT-4o Mini	One-shot	En→Tl	907	23.3%	6.2%	3.77	+153
GPT-4o Mini	One-shot	Tl→En	864	24.7%	5.9%	4.18	+160
GPT-4o Mini	One-shot	En→Es	693	17.5%	5.6%	3.10	+81
GPT-4o Mini	One-shot	Es→En	785	10.6%	4.6%	2.31	+46
GPT-4o Mini	Reason-then-edit	En→Tl	422	28.4%	7.6%	3.75	+87
GPT-4o Mini	Reason-then-edit	Tl→En	383	36.8%	8.9%	4.15	+106
GPT-4o Mini	Reason-then-edit	En→Es	164	28.0%	8.5%	3.29	+32
GPT-4o Mini	Reason-then-edit	Es→En	193	25.4%	7.3%	3.50	+35
Llama 3 8B	Zero-shot	En→Tl	821	11.6%	22.7%	0.51	−90
Llama 3 8B	Zero-shot	Tl→En	541	13.7%	14.8%	0.93	−6
Llama 3 8B	Zero-shot	En→Es	715	12.2%	13.4%	0.91	−9
Llama 3 8B	Zero-shot	Es→En	584	8.6%	11.5%	0.75	−17
Llama 3 8B	One-shot	En→Tl	895	12.7%	18.5%	0.69	−51
Llama 3 8B	One-shot	Tl→En	722	14.3%	11.5%	1.24	+20
Llama 3 8B	One-shot	En→Es	865	16.3%	11.1%	1.47	+44
Llama 3 8B	One-shot	Es→En	869	10.5%	9.9%	1.06	+5
Llama 3 8B	Reason-then-edit	En→Tl	931	9.8%	31.9%	0.31	−204
Llama 3 8B	Reason-then-edit	Tl→En	824	9.6%	30.5%	0.31	−170
Llama 3 8B	Reason-then-edit	En→Es	583	10.8%	23.3%	0.46	−72
Llama 3 8B	Reason-then-edit	Es→En	704	8.8%	25.3%	0.35	−115

Improved / harmed: native ΔCOMET > +0.02 / < −0.02. Unedited outputs have Δ = 0, so every improved or harmed item is an edited one.

Conditional on editing, the editors separate completely: the worst GPT-4o Mini harm rate (8.9%) is below the best Llama harm rate (9.9%). Among GPT-4o Mini strategies, zero-shot has the lowest harm rate among edits in every direction

(tied with one-shot in En→Es); one-shot accepts a similar or modestly higher harm rate for the largest net gain in three directions (tied in Es→En). Llama's improvement rate among edits (median 11%) is less than half of GPT-4o Mini's (median 25%), and reason-then-edit raises its harm rate among edits to 23–32%.

7. Metric concordance, strategy ranking and cost (X7–X9)

Metrics. Across the 24 conditions, ΔCOMET, ΔBLEU and ΔchrF++ agree in sign in 21 cases, including all 18 Holm-significant ones. Rank correlations: COMET–chrF++ ρ = 0.97, COMET–BLEU ρ = 0.94, BLEU–chrF++ ρ = 0.98. The three disagreements are all non-significant Llama conditions in which COMET is flat while both lexical metrics fall: Zero-shot En→Es (ΔCOMET +0.00, ΔBLEU −0.84, ΔchrF++ −0.65); One-shot Tl→En (ΔCOMET +0.03, ΔBLEU −2.34, ΔchrF++ −1.16); One-shot En→Es (ΔCOMET +0.18, ΔBLEU −0.73, ΔchrF++ −0.53). This is the signature of rewording away from the reference without measured semantic loss, consistent with the source review. BLEU and chrF++ remain descriptive; no tests were run on them.

Strategy ranking. Kendall's coefficient of concordance across the four directions is W = 1.00 for GPT-4o Mini (one-shot first, zero-shot second, reason-then-edit third in every direction) and W = 0.81 for Llama (one-shot and zero-shot swap in Es→En only). Reason-then-edit ranks last for both editors in all four directions. Rankings use point estimates; their significance is the F2 family's.

GPT-4o Mini strategy	Cost, all 4 directions (USD)	Prompt / completion tokens per item	Mean F1 Δ (points)	Δ points per US$ per 1,000 items
Zero-shot	0.294	334 / 38	+0.52	3.9–11.8
One-shot	0.384	482 / 38	+0.71	4.0–10.8
Reason-then-edit	0.417	449 / 60	+0.42	2.5–6.4

List uncached token rates from the Phase 4 run summary (US$1.10 for all 12,144 devtest calls; not an invoice). Llama ran locally on one A100; its tokens per item were 302/44 (zero-shot), 453/44 (one-shot) and 403/82 (reason-then-edit).

Cost. The retrieved exemplar is worth its tokens: one-shot costs 31% more than zero-shot for a 36% larger mean gain. Reason-then-edit is dominated: it costs 42% more than zero-shot, through both longer prompts and longer completions, and gains 19% less. At these prices, post-editing 1,000 sentences with GPT-4o Mini one-shot costs about US$0.09–0.10.

8. Final answers to the research questions

Confirmatory evidence first; exploratory refinements from this report are marked [X].

- **RQ1 (pairs, conditional on quality).** GPT-4o Mini gains are larger for Tagalog (+0.51 to +1.00) than for Spanish (+0.26 to +0.53), and Llama's losses are also largest for Tagalog. Editing helps weak drafts in 23 of 24 conditions in both pairs. [X] Quality composition explains 16–23% of GPT-4o Mini's pair gap; most of the rest arises in mid-quality drafts. No causal resource claim.
- **RQ2 (strategies).** Retrieved one-shot is GPT-4o Mini's best strategy (significant over zero-shot in three directions); reason-then-edit is never better and is Llama's worst in every direction. [X] The GPT-4o Mini ranking is identical in all four directions (W = 1.00); one-shot is also the most cost-effective; conclusions are unchanged under Bonferroni or joint Holm correction.
- **RQ3 (over-editing).** GPT-4o Mini over-edits 0.8–4.3% of strong drafts, Llama 11.2–20.9%. [X] Wilson intervals separate the pooled rates; conditional on editing, every GPT-4o Mini harm rate is below every Llama harm rate; for Llama, even sparing strong drafts would not remove the losses, because mid-quality drafts are also degraded.
- **RQ4 (direction).** Direction effects are small and inconsistent (into English larger for GPT-4o Mini in the Tagalog pair only). [X] The over-editing ordering is secure out of English but overlaps per direction into English. Reverse directions carry translationese; no direction claim beyond the estimates.

9. Phase 5 completion against the approved plan

Deliverable (Plan of 17 Oct, §4–6)	Status	Where
Full-precision metric JSON; signatures; COMET provenance	Done	docs/phase5/evidence/main_results.json, evaluation_spec.json
Native per-item COMET scores (16,973 unique triplets)	Done (private)	Drive eval_aaff72ad…/segment_scores_numeric.jsonl
Coverage audit: 24 × 1,012, aligned to S0	Done	Report 5, verification table
Bootstrap configuration, F1 and F2 Holm families	Done	F1_primary.csv, F2_strategy.csv; Report 5 §2–3
Effect tables, pool denominators, bands	Done	Report 5 §5–6; paper/generated/*
Over-editing with sensitivity 0.01 / 0.02 / 0.05	Done	Report 5 §5; this report §6
Failure summaries and preservation screens	Done	Report 5 §7 (0 fallbacks in all 24 conditions)
Runtime and token accounting	Done	Report 4 §5–6; this report §7
LaTeX results table (main_results_matrix.tex)	Done	paper/generated/, byte-identical to regeneration
Figures regenerated from records, never by hand	Done	paper/figures/, docs/phase5/figures_extended/
Cross-artifact numerical checks	Done	Report 5 (max difference 0); this report (decomposition 2e−16)
Source-based review of flagged cases	Done	96 cases; source_review_summary.json
Robustness and sensitivity (exploratory)	Done	This report; extended/extended_analyses.json
Transfer of verified results into Overleaf	Done	5 pages; 77 audited held-out claims
Review by the PI	With you	Reports 3.2, 4, 5 and 5.1

10. Manuscript and repository changes

Three manuscript edits follow from Sections 4–6; each new number is registered in *paper/claims_manifest.json* and recomputed by the audit from the committed evidence. The paper still compiles to exactly 5 pages including references.

Section	Before	After
Results, RQ1	“The pair difference thus tracks baseline quality as much as language.”	“Baseline quality explains only part of the pair difference, however: reweighting each direction to the pooled weak/middle/strong mix narrows GPT-4o Mini’s Tagalog–Spanish gap by 16–23%, and at least 73% of the remaining gap arises in mid-quality drafts (exploratory).”
Results, new paragraph	—	“Robustness (exploratory).” Multiplicity invariance of F1; sign agreement of the 18 significant effects across three metrics; Wilson separation of over-editing (≤ 5.6% vs ≥ 9.5%); harm among edited drafts (≤ 8.9% vs ≥ 9.9%).
Discussion, first paragraph	“A post-editor is therefore only useful if it leaves correct drafts alone.”	Adds: an oracle that left strong drafts untouched would keep Llama below S0 in 6 of 12 conditions, because it also degrades mid-quality drafts; protecting strong drafts is necessary but not sufficient.
Conclusion	“Gains came from weak drafts”	“Gains came from weak and mid-quality drafts rather than strong ones” (the decomposition shows both pools contribute).

Repository additions on phase4-main: *src/analysis/phase5_extended.py*, *tests/phase5/test_extended.py*, *docs/phase5/evidence/extended/* (JSON and three CSVs), *docs/phase5/figures_extended/*, this report, eleven new audited facts in *src/analysis/phase5_facts.py*, and updates to the reviewer memo and README. The public-file audit passes; no benchmark text or generations are added.

11. Limitations of the extended analyses

- Post hoc: these analyses were chosen after the devtest results were seen. They re-express or bound frozen results; they do not create new confirmatory claims.
- Aggregates only: three coarse quality pools cannot separate finer quality differences inside the middle pool. A segment-level standardization (for example, by S0 COMET deciles) would need the private segment file on Drive.
- Independence approximation: Newcombe intervals for editor differences ignore that both editors edit the same strong drafts.
- Oracle gates use the reference and bound headroom only; a deployable gate would need reference-free quality estimation, chosen on dev.
- Costs use list token rates recorded at run time, not invoices; local Llama compute is not priced.

12. Decisions requested

- **Author byline and order** (the manuscript still shows a working-draft byline).
- **S0 baseline-reproduction disposition** (Report 3.2 §5: 412/421 dev multi-sentence items identical; disclosed in the paper). Please acknowledge or advise.
- **Tagalog verification** of the 48 Tagalog-direction review cases (blind, three labels, about two hours), or keep it as a stated limitation.
- **The exploratory paragraphs and the RQ1 wording correction** in Results and Discussion: keep as written, shorten, or move to a footnote.

Evidence

Extended analyses ce3386e870f210a3fe0cdaf2a0db081aa5ef6b981d6c29cba44885932b82570c
Analysis script 93bc9a1982180b1c9a2692c8d67a15c2851ba13980793f6172f15e0fd67a88a2
Phase 5 results f2144b7e7c446fc63956c5e72825357bc1394ab24ac83c36ea0de1f96710d495
Evaluation aaff72ad3e0a432c7b12c341abe01f6439e37f12be5daf94579839a972e4e147
Freeze 6e6259a9cf1c7e2aca978a770fa85d86596f31b0e8b3c55f42635203e4f7462c